Generative Project

Text-To-Image Generation Using Transformer And Diffusion Models

Milestone 1

Group 02

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Git Repo Link: https://github.com/abdul6253/ie7615-group-02-project2

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1. Project Overview and Objectives

In Project 2 we build a system that makes a new picture from a short text. You type a caption like "A dog runs on the beach", and the system creates a 512×512 image of it.

The system has two parts. The **text encoder** (CLIP) reads the caption and turns it into numbers. The **image generator** (Stable Diffusion 1.5) uses those numbers to draw the picture.

This project is the opposite of Project 1. Project 1 was *discriminative*: it looked at a picture and predicted a label. Project 2 is *generative*: it starts from text and creates a new picture.

Goals for Milestone 1:

- Choose a dataset of photos with captions and prepare a clean set of 1,000 to 5,000 pairs.
- Set up the software on a GPU and show that CLIP turns captions into numbers correctly.
- Make our first images for 10 to 20 captions with a ready-made Stable Diffusion model, and compare them with the real photos.
- Write the project plan and set up the team GitHub repository.

2. Methodology

2.1 Dataset

We use **2,000 photos with captions** chosen at random from **Flickr30k** [1]. Flickr30k has 31,783 everyday photos, and each photo has 5 captions written by people.

We chose Flickr30k because people wrote the captions while looking at each photo, so the text matches the picture well. Other datasets, like Conceptual Captions and LAION, use text copied from websites, which often does not match. Flickr30k is also allowed for research and education. We download only the rows we need from Hugging Face.

Group	Pairs	Used for
Train	1,600	Training the model in later milestones (if we do it)
Validation	200	Choosing the best settings in Milestones 2 and 3
Test	200	Final scores and the image gallery (never used for choosing settings)

Table 1. How we split the 2,000 pairs (random, seed 42).

2.2 Preparing the data

We keep a photo and its caption only if they pass every check below. All settings are saved in a file (*dataset_config.json*), so anyone can build the exact same dataset again.

1. **Clean the captions:** fix extra spaces ("dog ." becomes "dog"), remove the final period, start with a capital letter.
2. **Check the captions:** keep 6 to 30 words; remove captions with a question mark (the writer was not sure), captions that are mostly symbols, and captions with words about nudity, violence or weapons.
3. **Pick one caption:** each photo has 5 captions; we keep the good one closest to 15 words.
4. **Check the photos:** remove photos that are too small (under 256 pixels) or very wide or tall.
5. **Remove repeats:** remove repeated captions, and photos that look almost the same as one we already have.
6. **Resize:** cut a square from the center, resize it to 512×512 pixels, and change the pixel values to the range -1 to 1 (the format the image generator needs).

2.3 Models

CLIP [2] learned from 400 million photos with text. It can turn a caption and a photo into lists of numbers (called *embeddings*) that are close together when they mean the same thing.

Stable Diffusion 1.5 [3] uses CLIP's text part. Figure 1 shows how it works: it starts from random noise and removes the noise step by step. At each step it looks at the caption's numbers, so the picture slowly becomes what

the caption says. It works on a small, compressed version of the image (called the *latent*), and a decoder turns the result into the final picture.

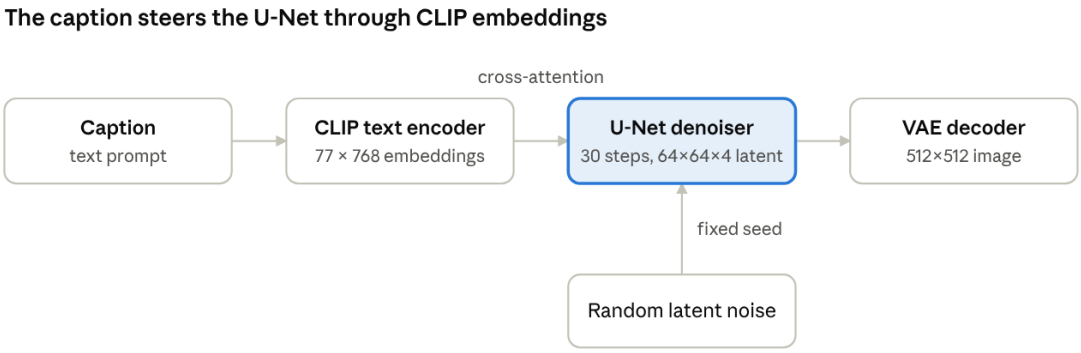


Figure 1. How the caption guides Stable Diffusion 1.5 to make an image.

2.4 Methods and tools

- **Tokenization:** CLIP splits each caption into small word pieces (always 77 pieces, with padding).
- **Similarity:** we measure how close a caption's numbers are to a photo's numbers (cosine similarity: 1 means a very good match).
- **Matching test:** for each caption, we check if CLIP picks its own photo out of many (Recall@K = how often the right photo is in the top K).
- **Making images:** 30 noise-removal steps, guidance scale 7.5 (how strongly the image follows the caption), and a fixed seed so results can be repeated.
- **CLIP score:** how well an image matches its caption (higher is better).
- **Tools:** Google Colab with an NVIDIA Tesla T4 GPU, Python 3.13, PyTorch 2.11, Hugging Face Transformers 5.20 and Diffusers 0.41. Exact versions are saved in *requirements.txt*.

3. Implementation and Results

We did the work in three notebooks in our GitHub repository: *01_dataset_preparation* (download, clean, resize and split the data), *02_clip_embeddings* (CLIP tokens, embeddings and the matching test) and *03_baseline_sd_generation* (images for 15 test captions, scores and gallery). Each notebook ends with automatic checks that stop with an error if something is wrong.

3.1 Dataset results

We checked 2,031 random photos to get 2,000 good pairs, so 98.5% of photos passed. Almost all removals were photos that were too small. No repeated captions or near-duplicate photos were found.

Step	Number of photos
Photos checked	2,031
Removed: photo too small (under 256 pixels)	30
Removed: photo too wide or tall	0
Removed: no good caption	1
Removed: repeated caption or photo	0
Kept	**2,000**

Table 2. Cleaning results for the photos.

Each photo has 5 captions, so we checked about 10,000 captions. About 95% passed. The rest were too short (378), too long (65), had unsafe words (34), showed the writer was unsure (6), or were mostly symbols (1). The captions we kept are 14 words long on average (shortest 7, longest 23), well under CLIP's limit.

Topic	People	Sports	Nature	Animals	City / street	Water / beach	Indoor
% of captions	70.2	15.5	13.4	12.2	10.4	7.8	5.8

Table 3. Topics mentioned in the captions (one caption can mention several).

Average pixel color (scale 0 to 1, 200 random photos): red 0.441, green 0.409, blue 0.371. After resizing, every photo is 512×512 with 3 color channels, and the pixel values go from -1 to 1 as Stable Diffusion needs.

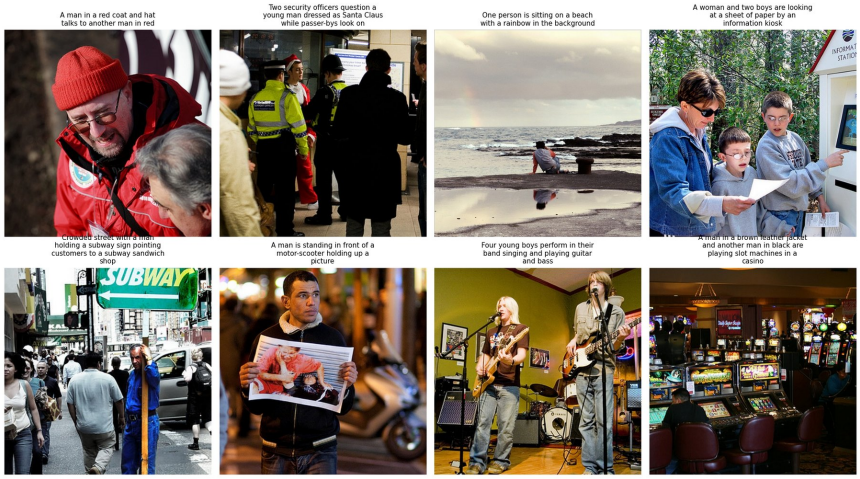


Figure 2. Examples of photos and captions after cleaning.

3.2 CLIP results

We tested CLIP on 500 random pairs. If it works, each caption should be much closer to its own photo than to a random photo.

Measure	Result
Average similarity, correct pairs	0.271
Average similarity, random pairs	0.097
Caption finds its photo: top 1 / top 5 / top 10	78.4 / 95.4 / 98.0 %
Photo finds its caption: top 1 / top 5	78.8 / 96.0 %
Random guessing, top 1	0.2 %

Table 4. CLIP matching test.

CLIP works very well on our data. Correct pairs are almost 3 times more similar than random pairs (0.271 vs. 0.097). For 78.4% of captions, CLIP's first choice out of 500 photos is the right one, about 390 times better than random guessing (0.2%), and the right photo is in the top 5 for 95.4% of captions. This shows the text→embedding pipeline runs correctly and gives meaningful numbers.

3.3 First generated images

We made one image for each of 15 test captions, about two for each scene type (people, animals, sports, water, city, nature, indoor). Each image took about 4.9 seconds on the T4 GPU, and the safety filter blocked none.

Measure	Result
Average CLIP score, generated images	27.0
Average CLIP score, real photos (same captions)	24.4
Average similarity, generated image vs. real photo	71.9
Similarity between 4 seeds (dog / dog in water / soccer)	93.0 / 88.1 / 81.8
Images blocked by the safety filter	0 of 15

Table 5. Scores for the first images (Stable Diffusion 1.5, PNDM scheduler, 30 steps, guidance 7.5).



Figure 3. Real photo (left) and generated image (right) for six of the 15 captions, with CLIP scores. Top row: successes. Bottom row: typical mistakes.

4. Analysis

4.1 What the results show

The pipeline works from start to end. CLIP finds the right photo for 78% of captions (Table 4), and Stable Diffusion made an image for all 15 test captions.

Generated images match the captions well. Their average CLIP score (27.0) is even higher than the real photos' (24.4). The model draws exactly what the caption says, clearly and in the center, while real photos often have extra objects, strange angles or bad light. But CLIP score uses the same kind of model that guides Stable Diffusion, so it may favor generated images; FID in Milestone 3 will test how real they look.

What worked well: animals and simple scenes (the dogs in #0, #1 and #9), and scenes with one strong idea, like the roller coaster (#12), the rave with glow sticks (#10) and the family by the Christmas tree (#8).

Common mistakes:

- **Counting:** "five people on bikes" gave three (#6), "three dogs" gave two (#9), and "three women and a man" gave only women (#11).
- **Numbers and text:** a "number ten jersey" shows "111" (#2).
- **Missing subjects in long captions:** "boy heads into water" shows only tents and hills (#5). This is one of only three captions where the real photo scores higher (25.1 vs. 22.7).
- **Negation:** "trees with no leaves" gave trees with leaves (#3).
- **Style and bodies:** generated images look brighter, like stock photos (similarity to the real photo only 71.9), and some bodies at the pool (#4) have strange shapes.

Generative vs. discriminative. One caption can describe many different pictures, so a generated image can match the caption well and still look very different from the real photo (#12 and #7 show this). That is why we judge images with the CLIP score and, later, FID, not by comparing pixels with one real photo.

Latent representation and conditional sampling. Stable Diffusion works on a small compressed version of the image (64×64×4, 48 times smaller than the 512×512 color image), which is why one image takes only about 5 seconds. Each image is one random sample "given" the caption; another seed gives another picture.

Mode collapse risk. Mode collapse means a model makes almost the same picture every time. In Figure 4, the four seeds give different poses, backgrounds and colors, so there is no mode collapse. But variety depends on the caption: the dog caption gave very similar images (similarity 93.0, always a tan dog on green grass), while the soccer caption was the most varied (81.8). In Milestone 2 we will test whether a lower guidance scale gives more variety.

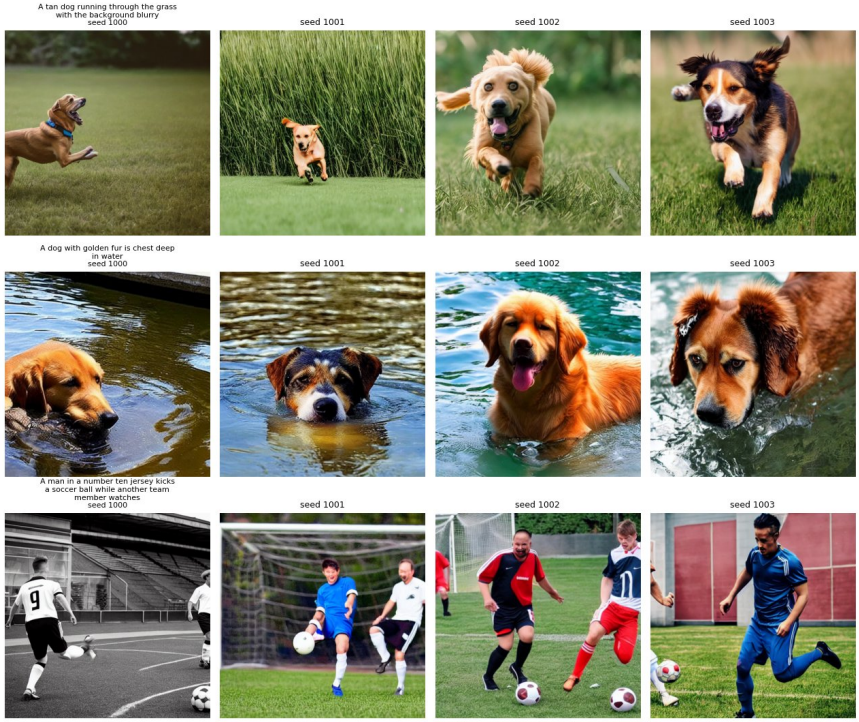


Figure 4. One caption, four different seeds (the mode collapse check).

4.2 Challenges

- **Limited GPU time:** free Colab can disconnect, so each notebook saves its results and can be run again in parts.
- **Dataset license:** the photos belong to their Flickr owners, so the 2,000 dataset photos are not stored on GitHub (only a few appear in our comparison figures); the notebook downloads them again.
- **Biased captions:** some captions guess a person's race or gender. We kept the original text but will point these out.

- **Small photos:** Flickr30k photos are at most 500 pixels, so making them 512 pixels enlarges them a little.

4.3 Ethics

The photos show real people who did not agree to AI research, so we use them only for comparison and never try to copy a real person. Stable Diffusion learned from internet images and can repeat stereotypes. We removed unsafe captions, keep the safety filter on, use all images only for this course, and will report biased results honestly.

4.4 Next steps (Milestone 2)

- Pass our own CLIP numbers into the image generator with our code, and compare CLIP with BERT as the text encoder.
- Test the guidance scale (1 to 15), three noise-removal methods (PNDM, DDIM, DPM-Solver++) and different numbers of steps.
- Prepare the FID and Inception Score code for Milestone 3.

5. Team Contributions

Member	Role	Contribution and responsibility
Abdellah Faleh	Technical lead	Built the three notebooks (data, CLIP, first images), ran them in Colab, and wrote the proposal, dataset memo, gallery memo and this report; connects the text encoder in Milestone 2
Jin-Woo Hong	Experiments	Leads the guidance scale, scheduler and step-count tests in Milestones 2 and 3
Samuel Tong	Evaluation	Leads the FID, Inception Score and CLIP score evaluation and the result charts in Milestone 3
Tristan Lyons	Documentation	Keeps the GitHub repository organized, submits on Canvas, and is the main contact with the teaching staff

References

[1] P. Young, A. Lai, M. Hodosh and J. Hockenmaier, "From image descriptions to visual denotations," *Transactions of the Association for Computational Linguistics*, vol. 2, pp. 67–78, 2014.

[2] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021.

[3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. CVPR*, 2022.

[4] J. Ho and T. Salimans, "Classifier-free diffusion guidance," *NeurIPS Workshop on Deep Generative Models*, 2021.